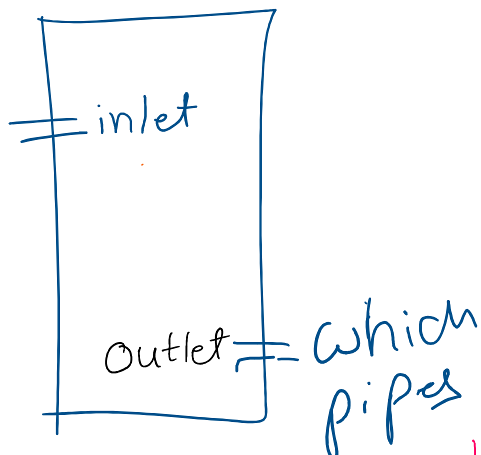
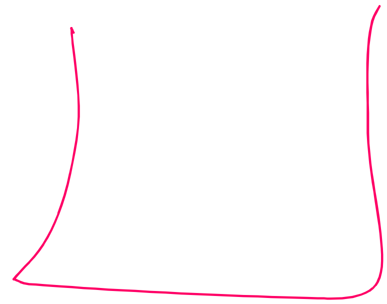
PIPES and CISTERN

Which pipes fills the tank



empty the tank



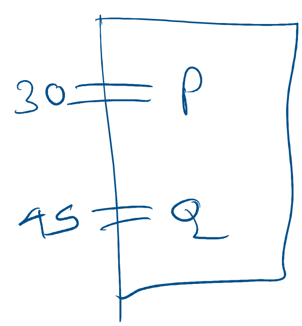
Cistern → open Storage

Two pipes P and Q can fill a tank in 30 hours and 45 hours respectively. If both the pipes are opened together, how much time will be taken to fill the tank?

P – 30, Q – 45 → 90: $\frac{90}{30} = 3$, $\frac{90}{45} = 2$

P+Q = 3+2 = 5 unit/hour

$\frac{90}{5} = 18$ hours



A pipe can fill a tank in 25 hrs. Due to a leakage in the bottom, it is

filled in 50 hrs. If the tank is full, how much time will the leak take to empty it?

total = inlet + outlet

total − inlet = outlet

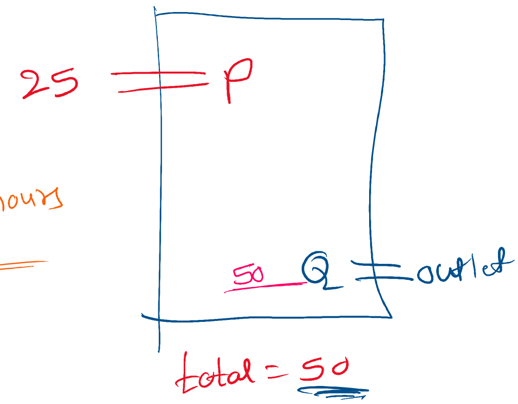
inlet ⇒ 25

total ⇒ 50

$50 \to \frac{50}{25} = 2$, $\frac{50}{50} = 1$

outlet ⇒ 2 − 1 = 1

$\frac{50}{1} = 50$ hours



Pipe P can fill a tank in 40 hours while Pipe Q alone can fill it in 50 hours and Pipe R can empty the full tank in 60 hours. If all the pipes are opened together, how much time will be needed to make the tank full?

P → 40

Q → 50

R → 60

$600 \to \frac{600}{40} = 15 +$, $\frac{600}{50} = 12 +$, $\frac{600}{60} = 10 -$

2 | 4, 5, 6
   2 5 3

4 × 5 × 3 = 60

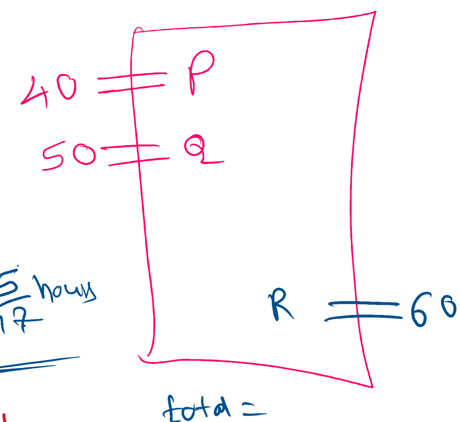
⇒ 15 + 12 − 10 ⇒ 17

17) 600 (35
  595
  (5)

$\frac{600}{17} \Rightarrow 35\frac{5}{17}$ hours

$\frac{3}{2}$ 2) 3 (1
  2
  (1)

$= 1\frac{1}{2}$ rem, div



Two pipes A and B can fill a cistern in 1.5 hour and 100 minutes respectively.There is also an outlet C. If all the three pipes are opened together, the tank is full in 60 minutes. How much time will be taken by C to empty the full tank?

total

by C to empty the full tank?

$A + B + C = \text{total}$

$A + B - \text{total} = C$

$A \Rightarrow 90$

$B \Rightarrow 100$

$\text{total} = 60$

900

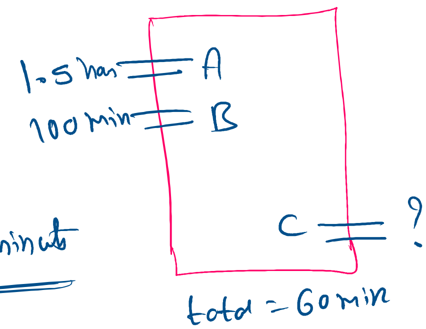
$\frac{900}{90} = 10 \quad +$

$\frac{900}{100} = 9 \quad +$

$\frac{900}{60} = 15 \quad -$

$10 + 9 - 15 = 4$

$\frac{900}{4} \Rightarrow 225$ minute



Two pipes A and B can fill a tank in 16 minutes and 24 minutes respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that that the tank is full in 12 minutes?

If two pipes function simultaneously, the reservoir is filled in 12 hrs. One pipe fills the reservoir 10 hrs faster than the other. How many hrs does the faster pipe take to fill the reservoir?

A tank has a leak which would empty it in 12 hrs. A tap is turned on which admits 6 litres a minutes into the tank, and it is now emptied in 16 hrs. How many litres does the tank hold?